Supplementary materials for "Topological nodal line semimetals with and without spin-orbital coupling"

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I. THE EXPLICIT Z_2 INVARIANT FOR A GAPPED AND REAL HAMILTONIAN IN 2D WITHOUT USING A SMOOTH GAUGE

In this Appendix, we explicitly construct the Z_2 invariant for a smooth mapping from a two-sphere to the manifold of all occupied states of a gapped and real Hamiltonian, $O(M+N)/O(M)\oplus O(N)$, without using smooth gauges on the northern and the southern hemispheres.

In this construction, we only need fix the gauges on the north pole and the south pole, where the wavefunctions of the occupied states are $|u_{NP,i}\rangle$ and $u_{SP,i}\rangle$ respectively, where i labels the occupied bands and both wavefunctions are real. For each azimuthal angle $\phi\in[0,2\pi)$, there is a half longitude connecting the two poles, $L(\phi)$. Consider the following path-ordered integral

$$W(\phi)=P\exp[\int_0^\pi dP(\theta,\phi)\partial_\theta P(\theta,\phi)d\theta], \tag{1}$$

or, in the discretized form

$$W(\phi)=\lim_{N\to\infty}\prod_{j=0,...,N}P(j\pi/N,\phi), \tag{2}$$

where $P(\theta,\phi)$ is the projector onto the occupied states, namely,

$$P(\theta,\phi)\equiv\sum_{i=1,...,N_{occ}}|u_i(\theta,\phi)\rangle\langle u_i(\theta,\phi)|. \tag{3}$$

The physical meaning of $W(\phi)$ is a parallel transport between the occupied space at the north pole and the occupied space at the south pole. Therefore, the following matrix must be unitary

$$M_{ij}(\phi)\equiv\langle u_{SP,i}|W(\phi)|u_{NP,j}\rangle. \tag{4}$$

Due to the reality of $|u_{NP,SP}\rangle$ and of the Hamiltonian, $M_{ij}(\phi)$ is also real and hence orthogonal. $M_{ij}(\phi)$ is hence a smooth mapping from a closed path to the space of orthogonal transforms. Due to $\pi_1[O(N)]=Z_2$, $M_{ij}(\phi)$ can either be trivial or nontrivial, and this Z_2 invariant of $M(\phi)$ is the Z_2 invariant of $H(\theta,\phi)$.

Given a matrix function $M(\phi)$ in $O(N_{occ})$, how do we know if it is trivial or nontrivial? Consider a unit vector in the real vector space of N_{occ} dimensions: $\mathbf{n}_0=(0,0,...,1)^T$. Acting $M(\phi)$ on this vector results in a path in $S^{N_{occ}-1}$:

$$\mathbf{n}(\phi)=M(\phi)\mathbf{n}_0. \tag{5}$$

Define

$$\begin{aligned}\mathbf{n}'(\phi)&\equiv\frac{\mathbf{n}(\phi)-(\mathbf{n}(\phi)\cdot\mathbf{n}_0)\mathbf{n}_0}{|\mathbf{n}(\phi)-(\mathbf{n}(\phi)\cdot\mathbf{n}_0)\mathbf{n}_0|},\\ \alpha(\phi)&=\cos^{-1}[\mathbf{n}_0\cdot\mathbf{n}(\phi)].\end{aligned} \tag{6}$$

Here $\mathbf{n}'(\phi)$ is the unit vector that is perpendicular to $\mathbf{n}_0$ and inside the plane spanned by $\mathbf{n}_0$ and $\mathbf{n}(\phi)$, and $\alpha(\phi)$ is the angle between $\mathbf{n}(\phi)$ and $\mathbf{n}_0$. Define the rotation

$$\begin{aligned}R(\phi,t)&=\cos[\alpha(\phi)t]+\sin[\alpha(\phi)t]\mathbf{n}_0\mathbf{n}'^T(\phi)\\ &-\sin[\alpha(\phi)t]\mathbf{n}'(\phi)\mathbf{n}_0^T,\end{aligned} \tag{7}$$

where $t\in[0,1]$. Under these definitions, the orthogonal matrix

$$O(\phi,t)=R(\phi,t)M(\phi), \tag{8}$$

is a smooth interpolation between $M(\phi)$ at $t=0$ and $M_0(\phi)$ at $t=1$ that satisfies

$$M_0(\phi)\mathbf{n}_0 = \mathbf{n}_0. \tag{9}$$

From Eq.(9), we see that $M_0(\phi)$ is an orthogonal matrix that is block diagonal: the entries in last row and in the last column are all zero except the unity on the diagonal. Therefore, the first $N_{occ}-1$ columns and rows of $M_0(\phi)$ is an orthogonal matrix in $N_{occ}-1$ dimensional real vector space. This block is defined as the new $M(\phi)$ and we repeat the process. On each step, we smoothly connect $M(\phi)$ to $M_0(\phi)$ that leaves the last unit vector invariant, then reduce the dimension of the matrix by one. This can proceed as long as $dim(M) > 2$, because the acting $M(\phi)$ on $\mathbf{n}_0$ gives a closed path in $S^{dim(M)-1}$, and since $\pi_1(S^{n>1}) = 0$, this path can always shrink to a point smoothly.

When $M(\phi)$ is reduced to an $O(2)$ matrix, we define the U(1) number

$$\beta(\phi) = M_{11}(\phi) + iM_{12}(\phi). \tag{10}$$

Due to the periodicity of $M(\phi)$, we can define the winding number of $\beta(\phi)$

$$2\pi n_w = -i\int \beta^*(\phi)\partial_\phi\beta(\phi)d\phi. \tag{11}$$

The Z_2 invariant is the parity of n_W, or,

$$\xi = \mathrm{mod}(n_w, 2). \tag{12}$$

II. THE Z_2 INVARIANT FOR EQ.(5)

The Z_2 invariant for any surface enclosing the origin can also be calculated for the model in Eq.(5). We will set $m = 0$ to calculate the invariant, and since it is a quantized, a small nonzero m does not change the result. Given a sphere around the origin, the smooth wavefunctions of the occupied bands on the northern and the southern hemispheres are

$$\begin{aligned}
|u_1^N\rangle &= (\cos\phi\sin\frac{\theta}{2}, -\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\sin\phi, 0)^T, \\
|u_2^N\rangle &= (\sin\frac{\theta}{2}\sin\phi, 0, -\sin\frac{\theta}{2}\cos\phi, \cos\frac{\theta}{2})^T, \\
|u_1^S\rangle &= (\sin\frac{\theta}{2}, -\cos\frac{\theta}{2}\cos\phi, 0, \cos\frac{\theta}{2}\sin\phi)^T, \\
|u_2^S\rangle &= (0, \cos\frac{\theta}{2}\sin\phi, -\sin\frac{\theta}{2}, \cos\frac{\theta}{2}\cos\phi)^T.
\end{aligned} \tag{13}$$

The connection matrix on the equator is found to be

$$M(\phi) = -\cos\phi - i\sin\phi\sigma_2. \tag{14}$$

This path corresponds to the nontrivial element of $\pi_1[O(2)]$ as the winding number is $+1$, hence the Z_2 invariant is nontrivial.

III. DOUBLE NODAL LINE IN A FOUR-BAND $k\cdot p$ MODEL WITH P, T AND A TWOFOLD SCREW AXIS

We revisit the 3D $k.p$-model in Eq.(7), in the presence of R_z in addition to $P*T$. When the origin of the $k.p$ expansion is at $\mathbf{k} = (0,0,0)$ ($\mathbf{k} = (0,0,\pi)$, we have $[R_z, P*T] = 0$ ($\{R_z, P*T\} = 0$). On At $\mathbf{k} = (0,0,0)$, the R_z is presented by $R_z = is_y$ up to a gauge. R_z puts constraints on the forms of $f_{i=1,2,3,4}$:

$$\begin{aligned}
f_{1,3}(q_x, q_y, q_z) &= -f_{1,3}(q_x, q_y, -q_z), \\
f_{2,4,5}(q_x, q_y, q_z) &= f_{2,4,5}(q_x, q_y, q_z).
\end{aligned} \tag{15}$$

Eqs.(15) guarantee that $f_{1,3}$ vanish on $q_z = 0$ plane. But it takes fine tuning to make $f_{2,4,5}$ vanish, i.e., to obtain a band crossing. This is consistent with our previous result that when $[R_z, P*T] = 0$, there is no symmetry protected

band crossing. If $\{R, P*T\}=0$, or at $\mathbf{k}=(0,0,\pi)$, R_z is represented by $R_z=is_x$ up to a gauge and the constraints on $f_{i=1,2,3,4,5}$ by R_z are

$$\begin{aligned} f_1(q_x,q_y,q_z) &= f_1(q_x,q_y,-q_z), \\ f_{2,3,4,5}(q_x,q_y,q_z) &= -f_{2,3,4,5}(q_x,q_y,-q_z). \end{aligned} \tag{16}$$

Eqs.(16) guarantee that $f_{2,3,4,5}$ vanish on $q_z=0$ plane. Therefore, in order to obtain a band crossing on this plane, we only need $f_1(q_x,q_y,0)=0$, the solution space to which is generically a 1D line, i.e., a double nodal line.

IV. TOPOLOGICAL INVARIANT FOR A DOUBLE NODAL LINE

According to the text, the double line crossing is formed by two doublet bands that have opposite eigenvalues of $R\equiv P*S_z$ (where S_z can be replaced by any twofold screw axis). For a given double nodal ring on the $k_z=\pi$-plane, we can find two points inside and outside the ring on the same plane, denoted by A and B. At A (B), each band is doubly degenerate, where the two degenerate states have the same eigenvalue of R, labeled by $\xi_n(A)$ ($\xi_n(B)$). We can hence define the following invariant

$$\zeta=\prod_{2n\in occ}\frac{\xi_{2n}(A)}{\xi_{2n}(B)}. \tag{17}$$

Since R only has eigenvalues of g_+ and $g_-=-g_+$, $\zeta=\pm1$ is a Z_2 invariant. When $\zeta=1$, the double nodal line between A and B is trivial and can be gapped, and if $\zeta=-1$, the line is protected.